Worksheet 1A: Fractions Fundamentals

Part A: Numerical Fractions (12 problems)

Simplify or evaluate. Show your work.

1. $\frac{3}{4} + \frac{5}{6}$

2. $\frac{7}{8} - \frac{2}{3}$

3. $\frac{5}{9} \times \frac{3}{10}$

4. $\frac{4}{5} \div \frac{2}{15}$

5. $\frac{2}{3} + \frac{1}{4} - \frac{1}{6}$

6. $\frac{7}{12} \times \frac{8}{21} \times \frac{3}{2}$

7. $\dfrac{\frac{3}{4}}{\frac{9}{16}}$

8. $\dfrac{\frac{2}{5} + \frac{1}{3}}{\frac{1}{2}}$

9. $2\frac{1}{3} \times 1\frac{1}{2}$

10. $4\frac{2}{5} \div 1\frac{1}{10}$

11. $\dfrac{5!}{4!}$

12. $\dfrac{7!}{5! \cdot 2!}$

Part B: Pattern Sums (8 problems)

Evaluate each sum. Look for patterns.

13. $1 + \frac{1}{2} + \frac{1}{4}$

14. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

15. $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

16. Based on problems 13–15, what value does this sum approach if we keep adding terms forever?

17. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4}$

(Hint: Try writing each fraction as a difference of two simpler fractions)

18. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5}$

19. Based on 17–18, if we kept adding terms $\frac{1}{n(n+1)}$ forever, what would the sum approach?

20. $1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!}$

(This sum, extended infinitely, equals $e \approx 2.718...$)

Part C: Algebraic Fractions — Simplification (10 problems)

Simplify completely. Assume no denominators equal zero.

21. $\frac{6p^2}{9p}$

22. $\frac{15m^3n^2}{25mn^4}$

23. $\frac{t^2 - 4}{t + 2}$

24. $\frac{r^2 - 9}{r^2 + 6r + 9}$

25. $\frac{2k^2 + 4k}{k^2 + 2k}$

26. $\frac{h^2 - 5h + 6}{h^2 - 4}$

27. $\frac{n!}{(n-1)!}$

28. $\frac{(n+1)!}{n!}$

29. $\frac{(n+2)!}{n! \cdot 2!}$

30. $\frac{6s^2 - s - 2}{2s^2 + 3s - 2}$

Part D: Algebraic Fractions — Operations (10 problems)

Perform the indicated operations and simplify.

31. $\dfrac{2}{p} + \dfrac{3}{q}$

32. $\dfrac{5}{t+1} - \dfrac{2}{t-1}$

33. $\dfrac{3}{m} + \dfrac{2}{n} - \dfrac{1}{mn}$

34. $\dfrac{r}{r+2} + \dfrac{2}{r-2}$

35. $\dfrac{1}{k-1} - \dfrac{1}{k+1}$

36. $\dfrac{2s}{s+3} \times \dfrac{s^2-9}{4s^2}$

37. $\dfrac{w^2-4}{w+5} \div \dfrac{w-2}{w^2+5w}$

38. $\dfrac{1}{u} + \dfrac{1}{u+1} + \dfrac{1}{u+2}$

39. $\dfrac{p^2-1}{p^2+2p} \cdot \dfrac{p^2+4p+4}{p^2+p-2} \div \dfrac{p+2}{p}$

40. If $\dfrac{1}{m} + \dfrac{1}{n} = \dfrac{1}{k}$, solve for k in terms of m and n.

Part E: Complex Fractions (6 problems)

Simplify each expression.

41. $\dfrac{\frac{1}{p} + \frac{1}{q}}{\frac{1}{p} - \frac{1}{q}}$

42. $\dfrac{\frac{2}{t+1} - \frac{1}{t}}{\frac{3}{t}}$

43. $\dfrac{1 - \frac{1}{n^2}}{1 + \frac{1}{n}}$

44. $\dfrac{\frac{m}{n} + \frac{n}{m}}{\frac{m^2-n^2}{mn}}$

45. $\dfrac{1}{1 + \frac{1}{1+\frac{1}{k}}}$

46. $\dfrac{\frac{1}{h} - \frac{1}{h+1}}{h}$

Part F: Repeated Multiplication — Shrinking vs Growing (6 problems)

47. Calculate each: - (a) $\frac{3}{4}$ - (b) $\left(\frac{3}{4}\right)^2$ - (c) $\left(\frac{3}{4}\right)^3$ - (d) $\left(\frac{3}{4}\right)^4$ - (e) What happens to $\left(\frac{3}{4}\right)^n$ as n gets very large?

48. Calculate each: - (a) $\frac{5}{4}$ - (b) $\left(\frac{5}{4}\right)^2$ - (c) $\left(\frac{5}{4}\right)^3$ - (d) $\left(\frac{5}{4}\right)^4$ - (e) What happens to $\left(\frac{5}{4}\right)^n$ as n gets very large?

49. Complete the rule: When you repeatedly multiply a fraction that is _____ than 1, the result approaches _____. When you repeatedly multiply a fraction that is _____ than 1, the result grows toward _____.

50. $\left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4$

51. $\left(\frac{9}{10}\right)^{10}$ — estimate first, then calculate. Is it closer to 0, 0.5, or 1?

52. $\left(\frac{99}{100}\right)^{100}$ — without calculating exactly, estimate whether this is closer to 0, 0.3, 0.5, or 1. Explain your reasoning.

Part G: Basic "What Happens When..." (8 problems)

No calculation required — describe the behavior.

53. Consider $\frac{n}{100}$ where $n > 0$. - (a) As n gets very large? - (b) As n approaches 0?

54. Consider $\frac{100}{n}$ where $n > 0$. - (a) As n gets very large? - (b) As n approaches 0?

55. For $\frac{k+1}{k}$ where $k > 0$: - (a) Rewrite as a sum of two fractions. - (b) As $k \to \infty$, what value does this approach?

56. For $\frac{t^2}{t^2+1}$ where $t > 0$: - (a) Calculate when $t = 1$, $t = 10$, $t = 100$. - (b) What value does this approach as $t \to \infty$? - (c) Will it ever actually reach that value?

57. For $\frac{1}{1+n}$ where $n > 0$: - (a) As $n \to 0$? - (b) As $n \to \infty$?

58. For $\frac{n}{n+1}$ where $n > 0$: - (a) As $n \to \infty$? - (b) Is the fraction always less than 1, equal to 1, or greater than 1?

59. You have fraction $\frac{a}{b}$ where $a, b > 0$. - (a) If you double both numerator and denominator, what happens to the value? - (b) If you add 1 to both numerator and denominator, does the value increase, decrease, or depend on something? *(Test with $\frac{1}{4}$ and $\frac{3}{4}$)*

60. For $\left(1 + \frac{1}{n}\right)$ where $n > 0$: - (a) Calculate when $n = 1$, $n = 10$, $n = 100$. - (b) What value does this approach as $n \to \infty$? - (c) Bonus: $\left(1 + \frac{1}{n}\right)^n$ approaches a famous number as $n \to \infty$. Any guesses?